IIIT Dharwad

Time Series Forecasting and Applications

Electricity Demand Forecasting

A Comparative Study of Statistical Time Series, Machine Learning, and Deep Learning Approaches

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Abstract

This report presents a comprehensive study on electricity demand forecasting for the All-India power grid, using two datasets of different temporal resolutions: a daily dataset spanning approximately three years (May 2023 – March 2026) and a high-frequency 15-minute interval dataset covering roughly one year (April 2025 – March 2026). We evaluate a structured progression of forecasting paradigms, ranging from naive baseline methods to classical statistical models (ARIMA, SARIMA, SARIMAX), autoregressive linear models, ensemble machine learning models (Random Forest and XGBoost), and lightweight deep learning architectures (DLinear and NLinear). All models are evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE) on a strictly chronological held-out test set. Our results demonstrate that ensemble machine learning models – particularly Random Forest and XGBoost – significantly outperform all other approaches on both datasets. Feature engineering, including lag features, rolling statistics, and cyclical temporal encodings, is shown to be a critical driver of model performance.

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1 Introduction

Accurate forecasting of electricity demand is a critical problem in modern power systems management. Reliable predictions enable generation planning, efficient load balancing, reduction of energy wastage, and maintenance of grid stability. Under-forecasting can lead to blackouts and supply shortfalls, while over-forecasting results in unnecessary spinning reserves and economic inefficiency.

This project approaches electricity demand forecasting at two distinct temporal resolutions. The **daily dataset** enables medium-to-long-term planning, capturing broad seasonal and calendar-driven trends over a three-year horizon. The **15-minute interval dataset** targets short-term intra-day forecasting, capturing fine-grained periodicity essential for real-time grid operations.

A key motivation for this dual-resolution study is to understand how model effectiveness varies with data frequency, and whether the same modelling paradigms dominate across both scales. We follow a structured pipeline:

1. Exploratory Data Analysis (EDA)
2. Baseline forecasting and AR models
3. Classical time series models (ARIMA, SARIMA, SARIMAX)
4. Machine learning models (Random Forest, XGBoost)
5. Deep learning models (DLinear, NLinear)
6. Final model selection, re-training and evaluation on unseen test data

The analysis of the daily and 15-minute datasets is conducted independently and reported separately to ensure clarity and avoid conflation of results across resolutions.

2 Datasets and Preprocessing

2.1 Dataset Descriptions

The dataset used in this project was collected through web scraping from publicly available electricity demand sources, specifically from the website: https://grid-india.in/en/. Data was programmatically collected using Python-based web scraping tools including requests and BeautifulSoup. Along with structured data extraction techniques, we systematically retrieved daily and high-frequency (15-minute interval) demand records. The raw data was then cleaned, formatted, and stored in CSV files for further analysis.

2.1.1 Daily Dataset

The daily dataset contains All-India peak electricity demand readings at a daily frequency, spanning from May 1, 2023, to March 31, 2026 – a total of 1,078 observations after removing one duplicate record. The target variable is `max_demand_met_mw`, representing the maximum demand met (in megawatts) on each calendar day.

Table 1: Summary Statistics – Daily Dataset

Statistic	**Value (MW)**
Count	1,078
Mean	213,281.97
Std. Dev.	14,687.36
Minimum	159,720.00
25th Percentile	204,299.50
Median	213,926.00
75th Percentile	222,998.00
Maximum	250,070.00

2.1.2 15-Minute Interval Dataset

The high-frequency dataset contains demand readings at 15-minute intervals, spanning from April 2, 2025, to March 31, 2026 – a total of 33,307 observations after feature engineering. The target variable is `demand_met_mw`.

Table 2: Summary Statistics – 15-Minute Dataset

Statistic	**Value (MW)**
Count	33,307
Mean	195,372.70
Std. Dev.	21,344.75
Minimum	128,473.00
25th Percentile	183,223.50
Median	198,853.00
75th Percentile	210,898.00
Maximum	245,068.00

2.2 Data Quality

Both datasets were inspected for missing values and duplicate records. The daily dataset contained one duplicate row, which was removed. No missing values were found in either dataset. Data types were verified and corrected where necessary.

2.3 Train-Validation-Test Split

A strictly chronological split was employed to prevent data leakage. No shuffling was performed at any stage. The split proportions were 70% training, 15% validation, and 15% test.

Table 3: Data Split Summary

Dataset	Split	Date Range	Size
Daily	Train	2023-05-01 → 2025-05-04	733
	Validation	2025-05-05 → 2025-10-23	157
	Test	2025-10-24 → 2026-03-31	158
15-Minute	Train	2025-04-02 → 2025-12-17	23,314
	Validation	2025-12-17 → 2026-02-07	4,996
	Test	2026-02-07 → 2026-03-31	4,997

2.4 Evaluation Metrics

All models are evaluated using three complementary metrics:

$$\text{MAE} = \frac{1}{n}\sum_{t=1}^{n}|y_t - \hat{y}_t| \tag{1}$$

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{t=1}^{n}(y_t - \hat{y}_t)^2} \tag{2}$$

$$\text{MAPE} = \frac{100\%}{n}\sum_{t=1}^{n}\left|\frac{y_t - \hat{y}_t}{y_t}\right| \tag{3}$$

MAE captures average absolute magnitude of error; RMSE penalises large errors more heavily; MAPE provides a scale-independent relative error. Together they give a complete picture of model performance.

3 Exploratory Data Analysis

3.1 Daily Dataset EDA

3.1.1 Trend and Seasonality

The daily demand series exhibits a steady upward trend over the three-year period, with average demand increasing from around 180,000 MW in mid-2023 to above 230,000 MW by mid-2024. The highest demand values are concentrated in summer 2024, with the peak of 250,070 MW recorded on May 30, 2024, reflecting increased cooling requirements.

Monthly aggregation indicates a bimodal seasonal pattern, with higher demand during January–February and May–June, and a clear dip in November (around 195,000 MW). This behavior can be attributed to seasonal cooling and heating effects.

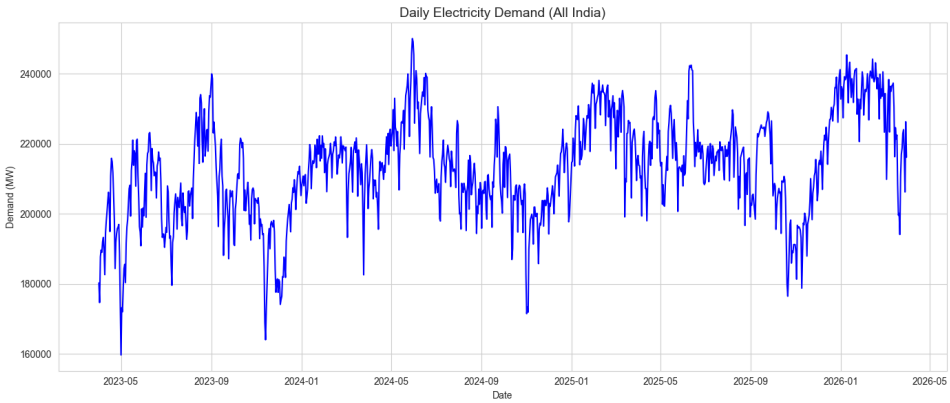

Figure 1: Daily Electricity Demand Over Time

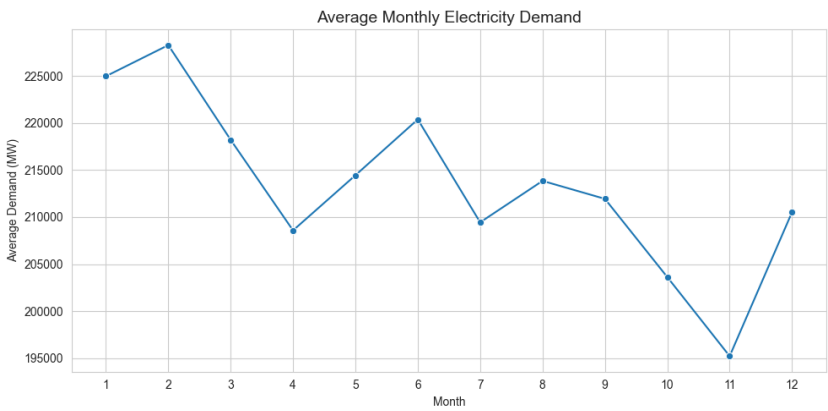

Figure 2: Average Monthly Electricity Demand

3.1.2 Weekly Patterns

A consistent weekly pattern is observed. Average weekday demand (214,929 MW) is higher than weekend demand (209,120 MW), indicating reduced activity during weekends.

Among weekdays, demand is highest on Fridays and lowest on Sundays.

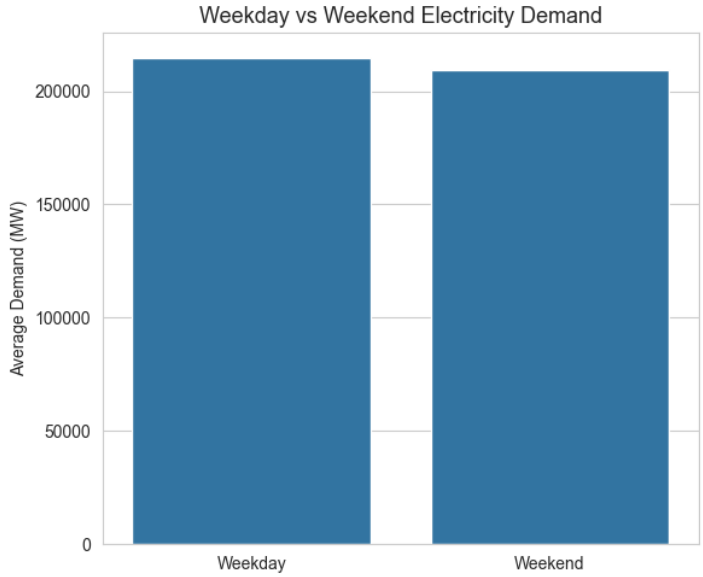

Figure 3: Weekday vs Weekend Demand

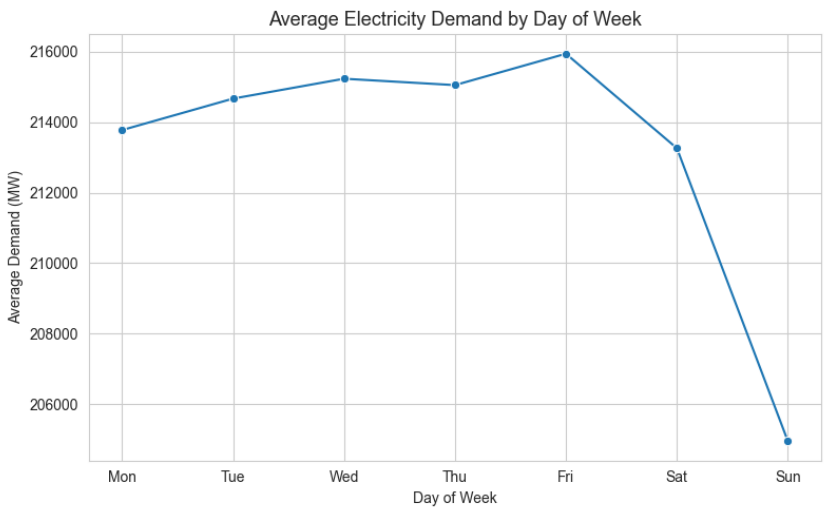

Figure 4: Average Demand by Day of Week

3.1.3 Festival Effects

Electricity demand on festival days is noticeably lower (207,248 MW) compared to regular days (213,906 MW). Pre- and post-festival periods show similar demand levels (around 212,600 MW), suggesting that the impact is primarily limited to the festival day.

The distribution of festival-day demand is slightly shifted toward lower values with greater variability.

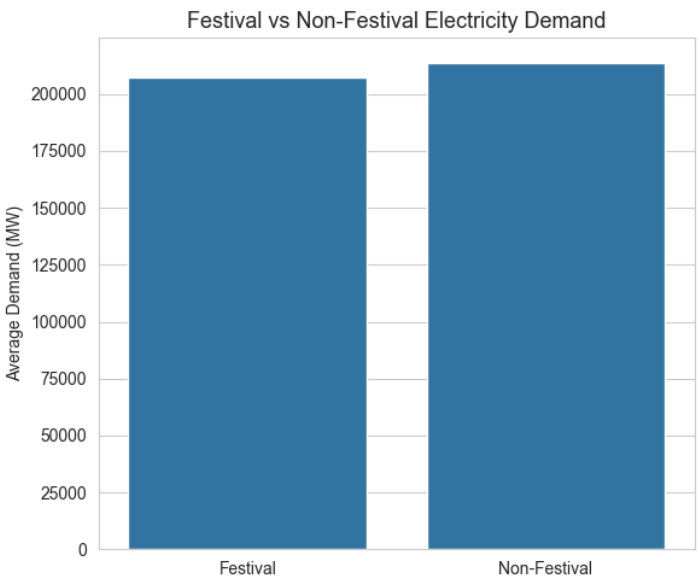

Figure 5: Festival vs Non-Festival Demand

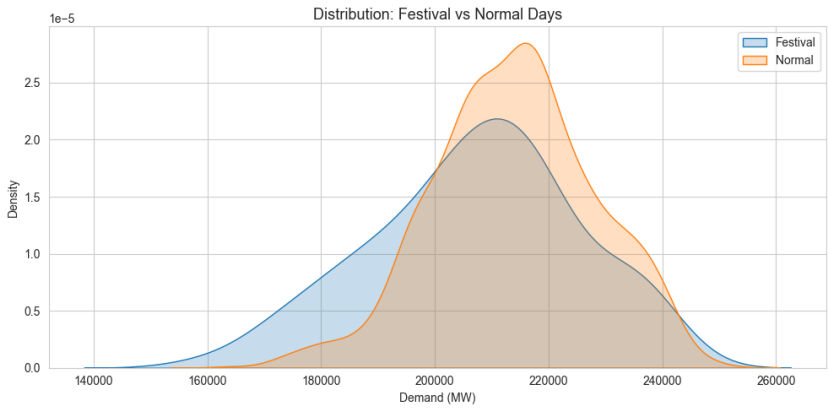

Figure 6: Distribution: Festival vs Normal Days

3.2 15-Minute Dataset EDA

3.2.1 Intra-Day Load Curve

The averaged intra-day load curve shows a clear double-peak pattern. Demand is relatively high around midnight (183,000 MW), drops to its lowest between 3:00–5:00 AM (174,000 MW), and then rises sharply during the morning hours.

The primary peak occurs around 10:30 AM (time block 42) with average demand near 212,200 MW, followed by a smaller secondary peak in the late afternoon. Demand gradually declines towards the night.

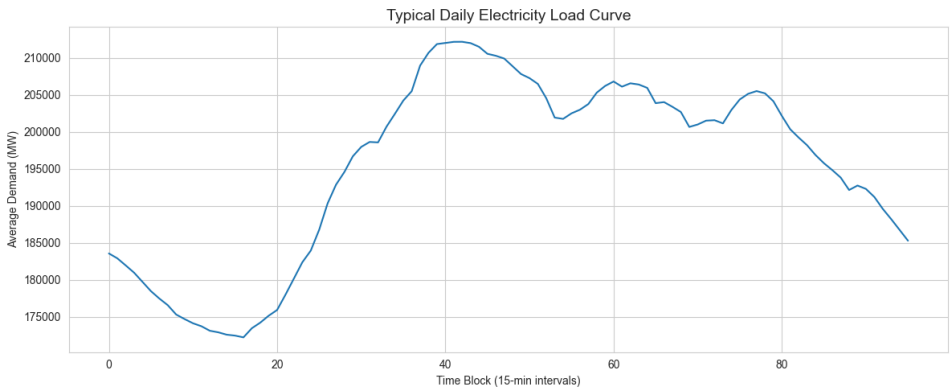

Figure 7: Typical Daily Load Curve

3.2.2 Festival vs. Normal Load Curves (15-Minute)

Comparative analysis shows that demand remains consistently lower on festival days across most time blocks, particularly during peak afternoon hours (time blocks 40–70). The average reduction is about 1,423 MW, with the largest drop (4,300 MW) observed during peak periods. In contrast, early morning demand is slightly higher on festival days, indicating a shift in usage patterns.

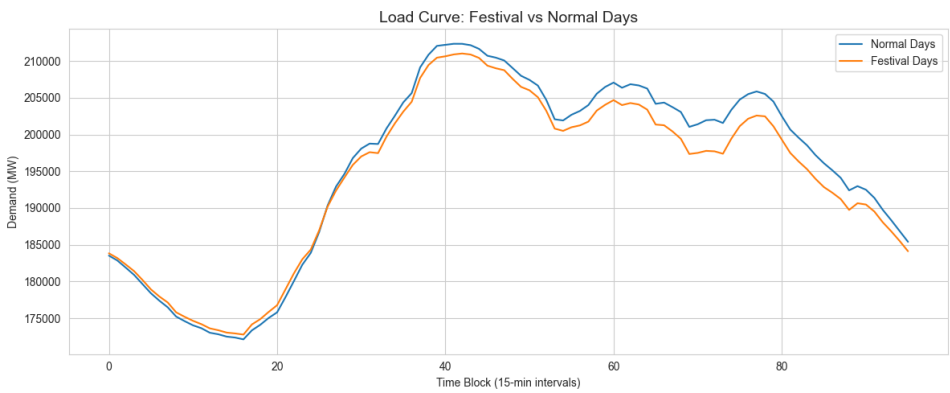

Figure 8: Load Curve: Festival vs Normal Days

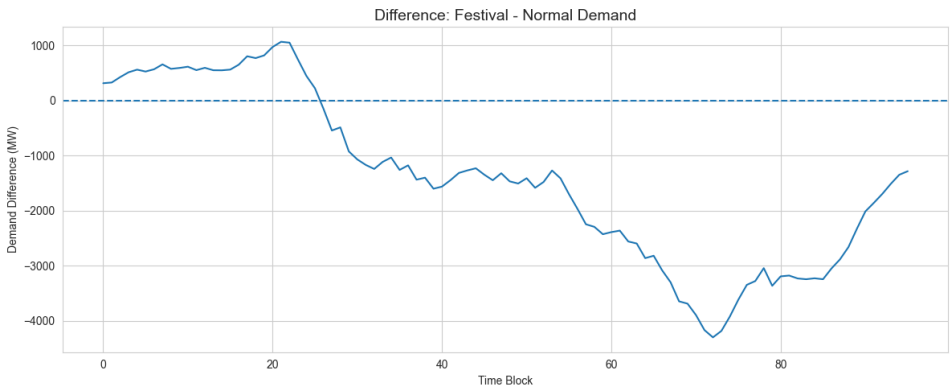

Figure 9: Difference: Festival - Normal Demand

4 Feature Engineering

4.1 Daily Dataset Features

Table 4: Features Engineered for the Daily Dataset

Feature	Description
`day_of_week`	Day of the week (0=Monday, 6=Sunday)
`month`	Calendar month (1–12)
`is_weekend`	Binary flag: 1 if Saturday or Sunday
`dow_sin`, `dow_cos`	Cyclical sine/cosine encoding of day-of-week
`month_sin`, `month_cos`	Cyclical sine/cosine encoding of month
`is_festival`	Binary: festival day indicator
`is_pre`, `is_post`	Binary: day before/after a festival
`lag_1`	Demand one day prior
`lag_7`	Demand seven days prior (same day, last week)
`lag_30`	Demand 30 days prior
`rolling_mean_7`	7-day rolling average of demand
`rolling_std_7`	7-day rolling standard deviation

4.2 15-Minute Dataset Features

Table 5: Features Engineered for the 15-Minute Dataset

Feature	Description
`hour`, `minute`	Hour and minute of the reading
`time_block`	Integer index 0–95 (position in the day)
`hour_sin`, `hour_cos`	Cyclical encoding of hour
`day_of_week`	Day of the week
`lag_1`	Demand at the immediately preceding 15-min interval
`lag_96`	Demand at the same time 24 hours prior
`is_festival`, `is_pre`, `is_post`	Festival indicators (merged from daily dataset)

5 Daily Dataset: Forecasting Models

This section summarises the models applied to the daily dataset, progressing from simple baselines to more advanced approaches.

5.1 Baseline Models

Baseline models establish reference performance levels. Any model that fails to exceed these provides limited practical value.

5.1.1 Mean Model

Predicts a constant value equal to the training mean. It does not incorporate any temporal structure.

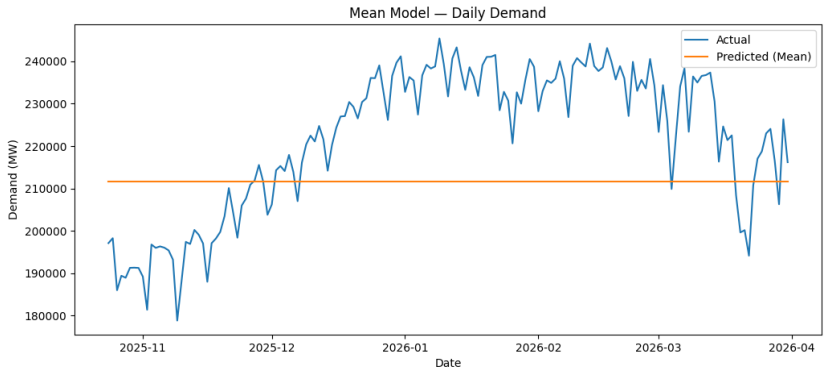

Figure 10: Mean Model Prediction vs Actual

5.1.2 Naive Model

Uses the previous day's demand as the prediction. This serves as a strong benchmark for time series with high autocorrelation.

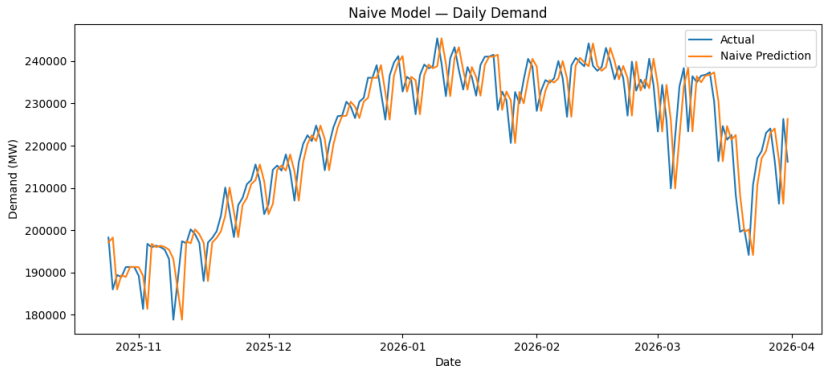

Figure 11: Naive Model Prediction vs Actual

5.1.3 Seasonal Naive Model

Predicts demand using the value from the same day in the previous week, capturing weekly periodicity.

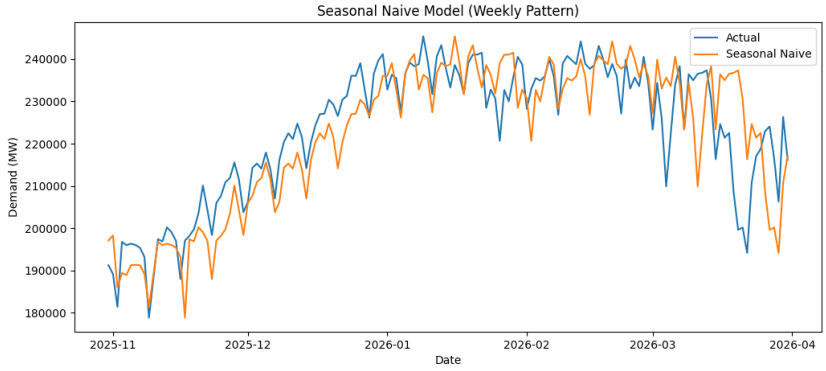

Figure 12: Seasonal Naive Model

5.1.4 Drift Model

Projects a linear trend between the first and last training observations and extends it into the future.

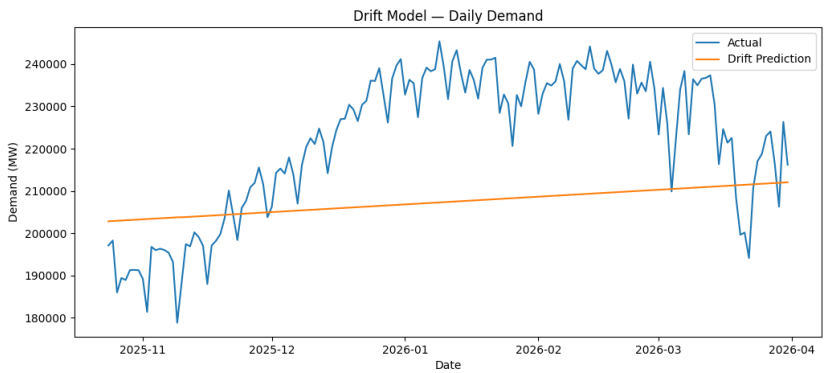

Figure 13: Drift Model Prediction

5.1.5 Simple Exponential Smoothing (SES)

Assigns exponentially decreasing weights to past observations, producing a smoothed forecast.

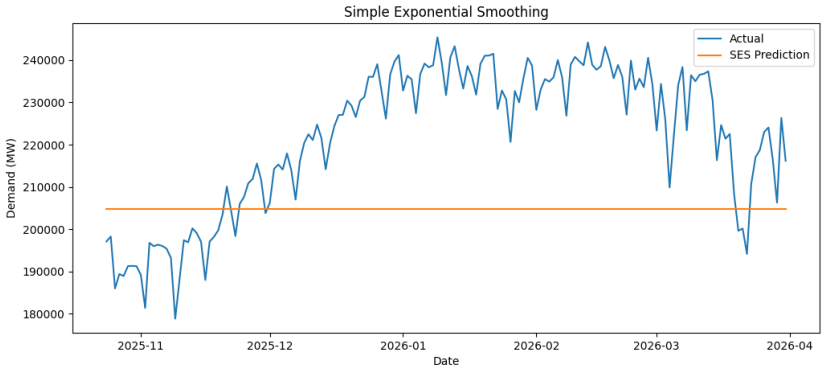

Figure 14: Simple Exponential Smoothing

5.1.6 Holt Linear Trend Model

Extends SES by including a trend component, allowing the forecast to adapt to increasing or decreasing patterns.

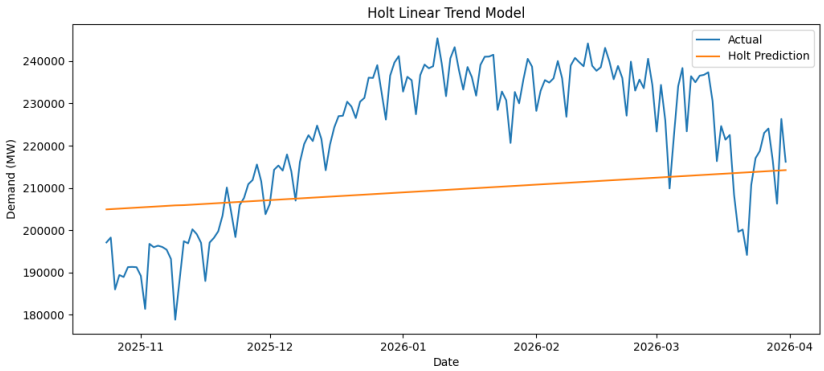

Figure 15: Holt Linear Trend Model

5.1.7 Holt-Winters Model

Further extends Holt's method by incorporating weekly seasonality in addition to trend and level.

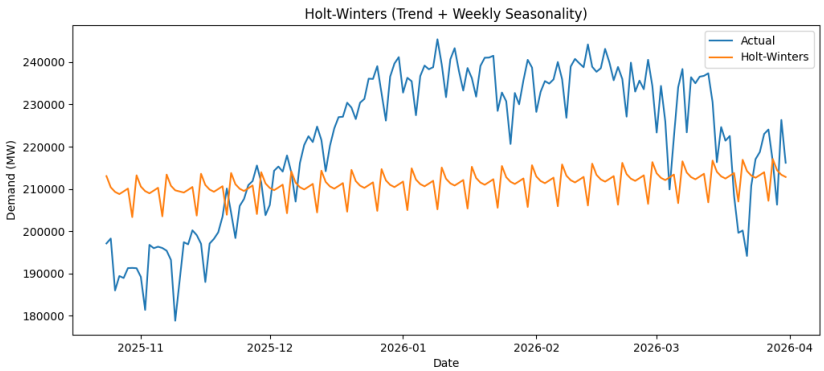

Figure 16: Holt-Winters Model

Key Insight

Among all baseline approaches, the Naive model performs best, indicating strong short-term dependency in the demand series. Smoothing-based methods tend to lag behind rapid changes and fail to capture variability effectively.

5.2 Autoregressive (AR) Linear Model

An autoregressive linear regression model was trained using lag features from the previous day and previous week.

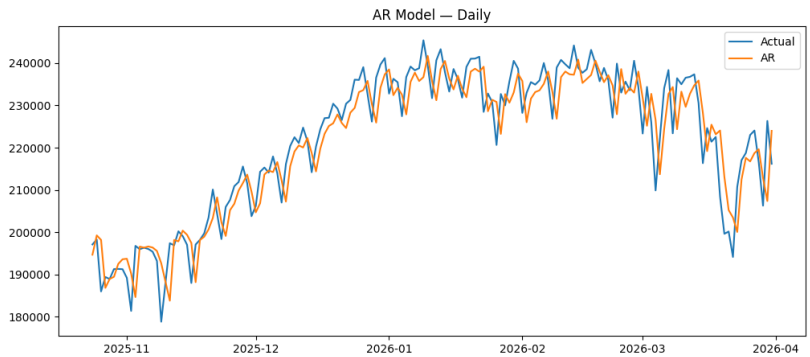

Figure 17: AR Model Prediction vs Actual

Key Insight

The AR model improves upon the Naive baseline by incorporating both short-term and weekly dependencies, confirming the importance of seasonal lag information.

5.3 Classical Time Series Models

5.3.1 ARIMA

ARIMA captures autoregressive and moving average components along with differencing for stationarity. However, it struggles with complex seasonal and external effects.

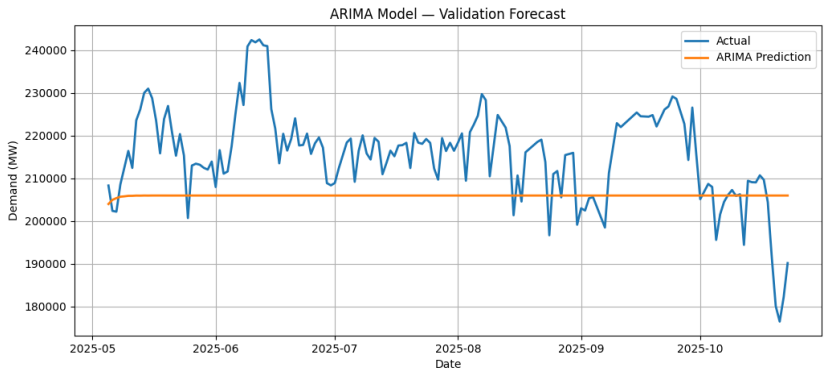

Figure 18: ARIMA Validation Forecast

5.3.2 SARIMA

SARIMA extends ARIMA by incorporating weekly seasonality, improving performance compared to ARIMA.

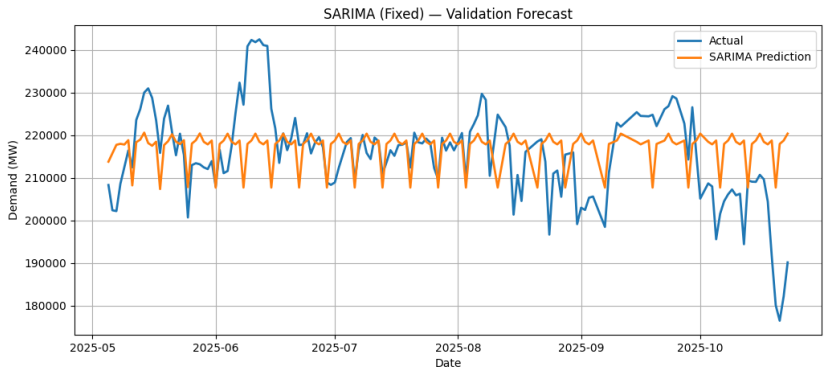

Figure 19: SARIMA Validation Forecast

5.3.3 SARIMAX

SARIMAX includes external variables such as festival and weekend indicators.

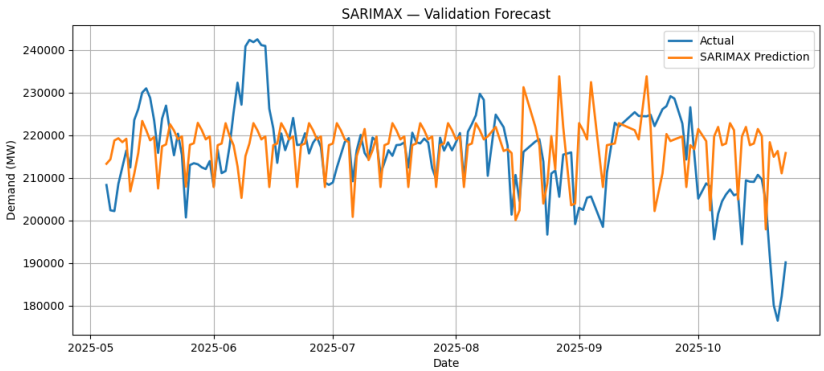

Figure 20: SARIMAX Validation Forecast

Key Insight

While SARIMA captures seasonality reasonably well, it still underperforms simpler baselines. Including exogenous variables in SARIMAX does not significantly improve results, indicating limited additional signal from these features.

5.4 Machine Learning Models

Machine learning models leverage engineered features to capture complex and non-linear relationships.

5.4.1 XGBoost

An ensemble boosting model that sequentially reduces prediction errors.

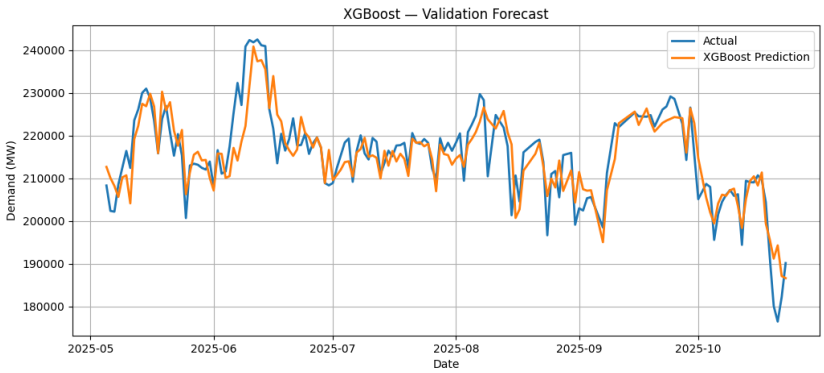

Figure 21: XGBoost Validation Forecast

5.4.2 Random Forest

An ensemble of decision trees trained using bagging, offering strong robustness and stability.

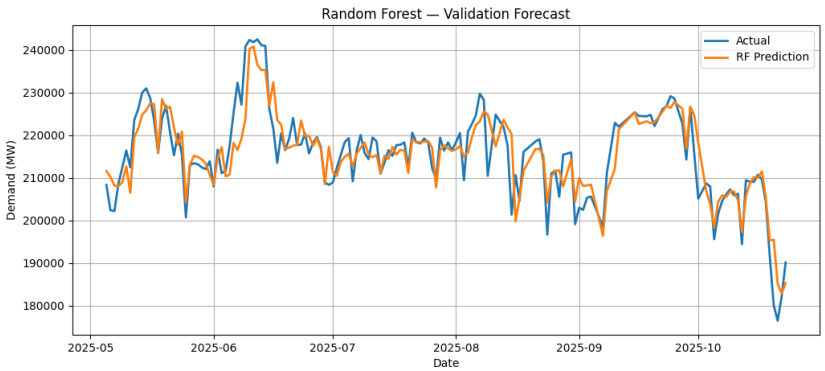

Figure 22: Random Forest Validation Forecast

Key Insight

Both models significantly outperform statistical approaches. Random Forest achieves the best performance, indicating that non-linear feature interactions and engineered features are critical for accurate demand forecasting.

5.5 Deep Learning Models

Lightweight linear neural models were applied using sliding window inputs.

Key Insight

Deep learning models perform poorly on this dataset. The limited data size, minimal training, and absence of engineered features restrict their ability to generalise, leading to significantly higher errors compared to simpler models.

5.6 Final Model Selection and Test Evaluation

Based on validation set performance, Random Forest and XGBoost were selected as the final models for the daily dataset. Prior to final evaluation, both models were retrained on the combined train+validation data (890 observations, covering May 2023 – October 2025) to maximise the volume of information available for learning.

Table 6: Final Test Set Performance – Daily Dataset

Model	MAE (MW)	RMSE (MW)	MAPE (%)
Random Forest	**4,151.56**	**5,377.17**	**1.88**
XGBoost	4,386.03	5,646.74	1.97

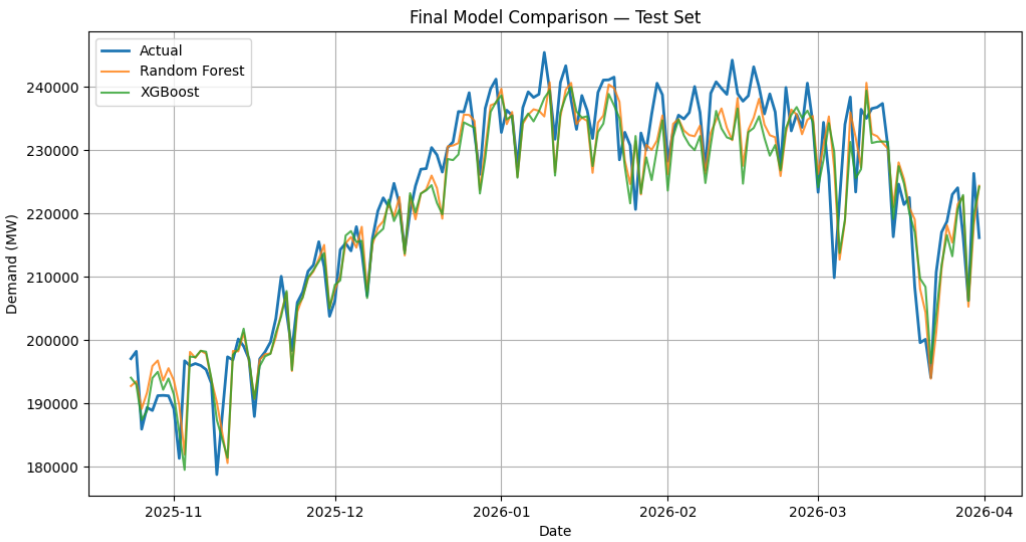

Figure 23: Final Model Comparison - Test Set

Key Insight

Random Forest emerges as the most reliable model, achieving the lowest prediction error. Both models generalise well, with minimal performance degradation on the test set, indicating stable learning without overfitting.

5.7 Daily Dataset: Model Comparison Summary

Model	Category	MAE (MW)	RMSE (MW)	MAPE (%)
Mean Model	Baseline	17,836	19,872	7.94
Drift Model	Baseline	18,731	21,346	8.20
Holt Linear	Baseline	17,612	19,930	7.75
Holt-Winters	Baseline	17,698	19,964	7.85
Seasonal Naive	Baseline	6,802	9,205	3.10
Naive Model	Baseline	5,174	6,756	2.36
AR (lag_1, lag_7)	Linear AR	4,936	6,187	2.24
ARIMA (val)	Classical TS	11,809	14,324	5.39
SARIMA (val)	Classical TS	8,133	11,131	3.86
SARIMAX (val)	Classical TS	8,526	11,485	4.03
DLinear	Deep Learning	31,558	32,767	14.00
NLinear	Deep Learning	223,051	223,834	99.14
XGBoost	Machine Learning	4,386	5,647	1.97
Random Forest	**Machine Learning**	**4,152**	**5,377**	**1.88**

6 15-Minute Dataset: Forecasting Models

This section summarises the models applied to the high-frequency 15-minute dataset, progressing from simple baselines to more advanced approaches.

6.1 Baseline Models

Baseline models establish reference performance levels. Any model that fails to exceed these provides limited practical value.

6.1.1 Naive Model

Uses the previous 15-minute demand value as the prediction. This serves as a strong benchmark for high-frequency time series with strong autocorrelation.

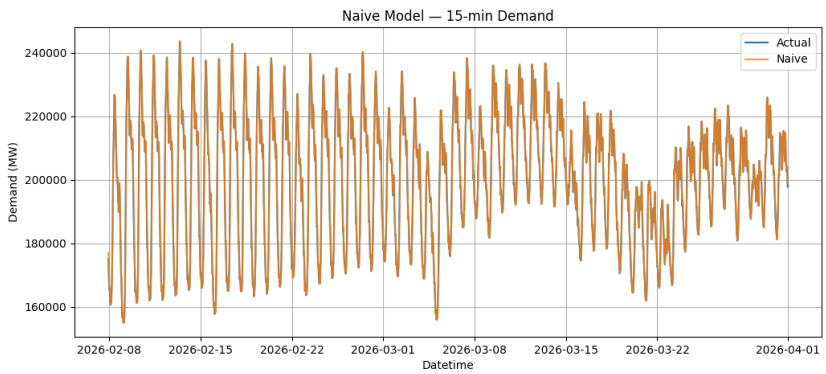

Figure 24: Naive Model — 15-Minute Demand

6.1.2 Seasonal Naive Model

Predicts demand using the value from the same time slot on the previous day (lag 96), capturing daily periodicity.

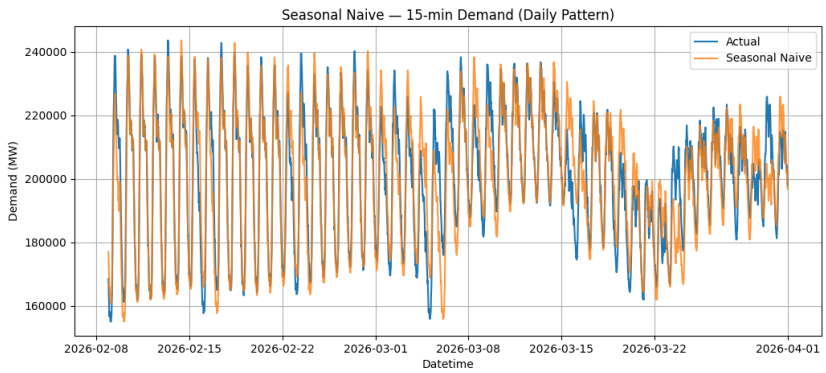

Figure 25: Seasonal Naive Model (Daily Pattern)

6.1.3 Rolling Mean Model

Predicts using the average of the last four observations (approximately one hour), providing a smoothed estimate.

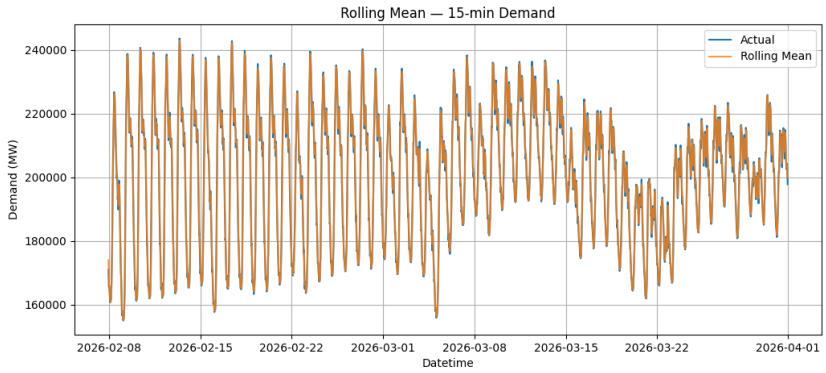

Figure 26: Rolling Mean Model

Key Insight

At 15-minute resolution, the Naive model performs best among baselines, indicating extremely strong short-term dependency. The Seasonal Naive model performs significantly worse, suggesting that intra-day patterns vary across days. Immediate past values dominate over historical daily patterns.

6.2 Autoregressive (AR) Linear Model

An autoregressive linear regression model was trained using lag features from the previous interval and the same time on the previous day.

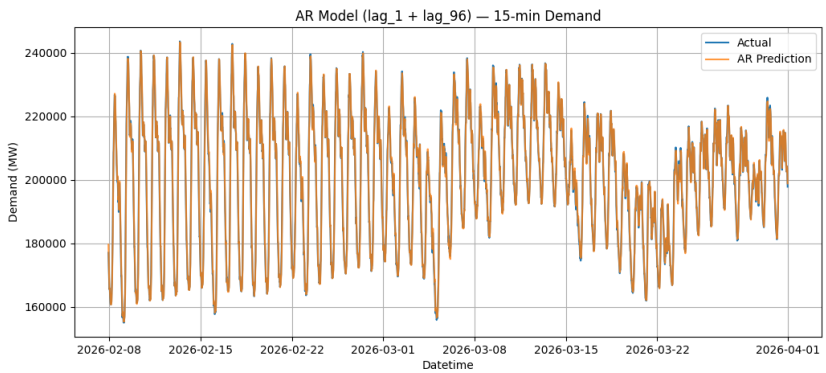

Figure 27: AR Model — 15-Minute Prediction vs Actual

Key Insight

The AR model provides only marginal improvement over the Naive baseline. While the daily seasonal lag contributes slightly, the immediate previous value remains the dominant predictor in high-frequency data.

6.3 Machine Learning Models

Machine learning models leverage engineered features to capture complex and non-linear relationships.

6.3.1 XGBoost

An ensemble boosting model that sequentially reduces prediction errors.

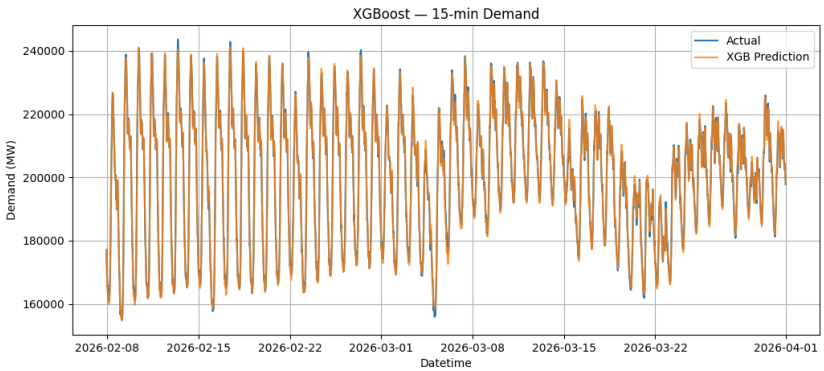

Figure 28: XGBoost — 15-Minute Validation Forecast

6.3.2 Random Forest

An ensemble of decision trees trained using bagging, providing robustness and stability.

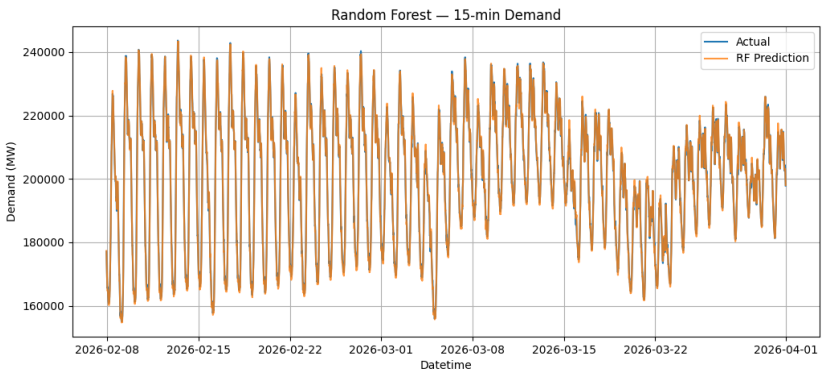

Figure 29: Random Forest — 15-Minute Validation Forecast

Key Insight

Both models significantly outperform baseline and linear approaches. The improvement is more pronounced than in the daily dataset, indicating that non-linear interactions between lag features and time-based features are more important at higher frequency. Random Forest and XGBoost perform very similarly.

6.4 Deep Learning Models

Lightweight linear neural models were applied using a sliding window of past observations.

6.4.1 DLinear

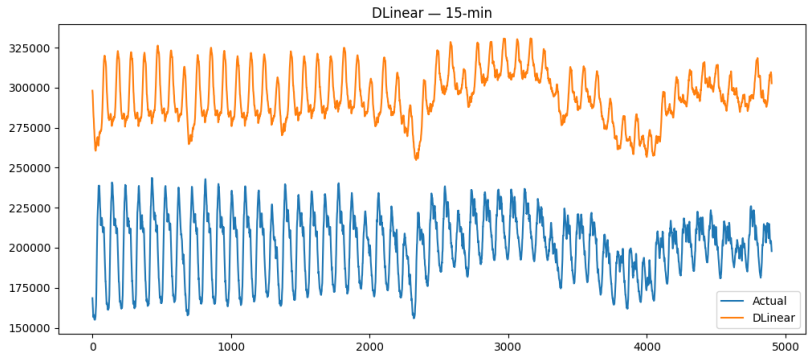

Figure 30: DLinear — 15-Minute Forecast

6.4.2 NLinear

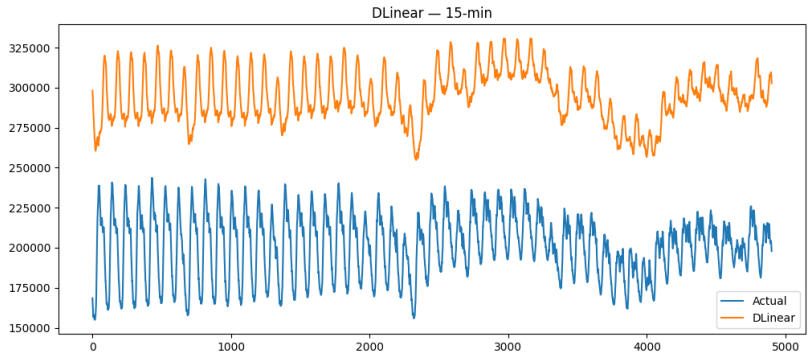

Figure 31: NLinear — 15-Minute Forecast

Key Insight

Deep learning models perform poorly on this dataset. Similar to the daily case, the lack of sufficient training, limited feature context, and simplistic architectures result in large prediction errors. These models fail to capture the strong short-term dynamics effectively.

6.5 Final Model Selection and Test Evaluation

Based on validation performance, Random Forest and XGBoost were selected for final evaluation on the 15-minute test set. Both were retrained on the combined train+validation data using the cyclically encoded feature set (`lag_1`, `lag_96`, `hour_sin`, `hour_cos`).

Table 7: Final Test Set Performance – 15-Minute Dataset

Model	MAE (MW)	RMSE (MW)	MAPE (%)
Random Forest	1,159.71	1,526.38	0.58
XGBoost	**1,121.41**	**1,463.99**	**0.56**

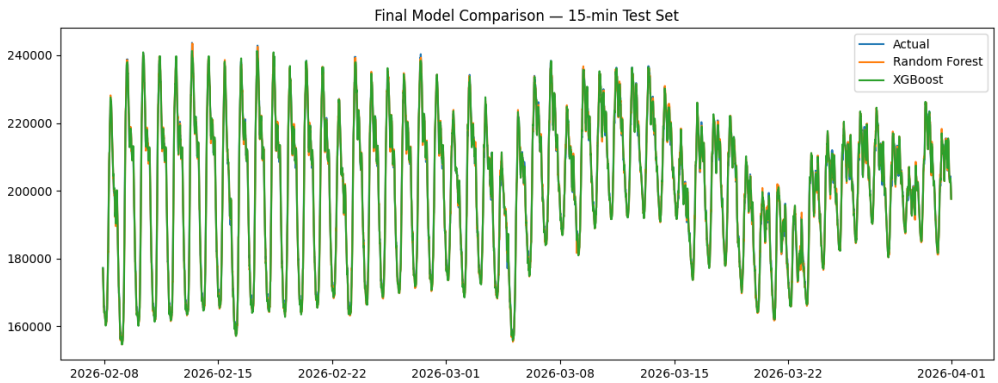

Figure 32: Final Model Comparison - Test Set

Key Insight

Random Forest emerges as the most reliable model for the 15-minute dataset, achieving the lowest prediction error. Both models generalise well, with stable performance on unseen data and minimal overfitting.

6.6 15-Minute Dataset: Model Comparison Summary

Table 8: Complete Model Comparison – 15-Minute Dataset

Model	Category	MAE (MW)	RMSE (MW)	MAPE (%)
Seasonal Naive (lag_96)	Baseline	5,709.08	7,969.66	2.86
Rolling Mean (4 steps)	Baseline	2,070.74	2,617.51	1.03
Naive (lag_1)	Baseline	1,521.77	1,962.43	0.76
AR (lag_1, lag_96)	Linear AR	1,486.38	1,913.29	0.74
DLinear	Deep Learning	15,354.67	19,228.75	8.07
NLinear	Deep Learning	200,757.10	200,949.42	100.94
Random Forest	Machine Learning	1,159.71	1,526.38	0.58
XGBoost	**Machine Learning**	**1,121.41**	**1,463.99**	**0.56**

7 Discussion

7.1 Cross-Dataset Comparison

The final model results reveal an interesting asymmetry between the two datasets. On the daily dataset, Random Forest (MAPE 1.88%) narrowly outperforms XGBoost (MAPE 1.97%). On the 15-minute dataset, the ordering reverses: XGBoost (MAPE 0.56%) narrowly outperforms Random Forest (MAPE 0.58%). In both cases, the margin is small, and both models substantially outperform all other approaches.

The absolute magnitude of error is notably lower for the 15-minute dataset (MAPE $< 1\%$) compared to the daily dataset (MAPE $\approx 2\%$). This can be attributed to the much higher autocorrelation at short intervals – consecutive 15-minute readings are highly similar – and to the richer intra-day structure that the lag and time features can exploit.

7.2 Why Machine Learning Models Dominate

The consistent superiority of ensemble ML models across both datasets and all evaluation metrics can be attributed to several factors:

1. **Non-linearity:** Decision tree-based ensembles can capture non-linear relationships between features and target without any functional form assumptions. Demand exhibits non-linear responses to calendar and seasonal effects.
2. **Feature interaction modelling:** Ensemble models implicitly capture interactions between features (e.g., the joint effect of being a Friday in summer versus a Friday in winter) that linear models cannot represent without explicit interaction terms.
3. **Robustness to outliers:** Tree-based models are naturally robust to outlier demand values caused by unusual weather or special events, whereas classical linear models can be distorted by such points.
4. **Rich feature engineering:** The availability of lag features, rolling statistics, and cyclical encodings provides the ML models with a rich representation of the temporal context, which is the core driver of their performance advantage.

7.3 Limitations of Classical Time Series Models

ARIMA and SARIMA are fundamentally limited to linear, stationary representations of time series. In the presence of complex seasonal patterns, non-linearities, and external drivers such as festivals, these models are insufficient. The addition of exogenous variables in SARIMAX did not resolve these structural limitations, as the model still captures only linear relationships.

7.4 Deep Learning Models: Caveats

The poor deep learning results should not be interpreted as an indication that neural networks are inappropriate for this problem. The architectures evaluated (DLinear, NLinear) were deliberately minimal and were trained under constrained conditions. Properly tuned recurrent networks (LSTM, GRU), attention-based models (Transformers, PatchTST), or modern foundation time series models would likely achieve competitive or superior performance, especially given sufficient data.

8 Conclusions

This project presents a systematic end-to-end comparative study of electricity demand forecasting across two temporal resolutions using a diverse range of modelling approaches. The key conclusions are as follows:

1. **Machine learning models are the most effective approach** for both daily and 15-minute forecasting. Random Forest and XGBoost consistently outperform all baselines, AR models, classical statistical models, and the deep learning architectures evaluated.
2. **Feature engineering is the most critical driver of performance.** The combination of lag features (capturing autocorrelation), rolling statistics (capturing local trends), and cyclical temporal encodings (capturing seasonality) enables ML models to represent the full temporal context of electricity demand.
3. **Classical time series models underperform even simple baselines.** ARIMA, SARIMA, and SARIMAX all fail to outperform the Naive model on the daily dataset, highlighting their inadequacy for complex, non-linear demand patterns.
4. **The Naive model is a strong and non-trivial baseline.** Any meaningful forecasting model must substantially beat the Naive model to justify deployment.
5. **High-frequency data yields lower relative errors.** The 15-minute models achieve MAPE below 1%, compared to approximately 2% for daily models, due to stronger short-term autocorrelation.
6. **Deep learning requires more investment to be competitive.** The minimal architectures evaluated here performed poorly, but this reflects the experimental setup rather than an inherent limitation of neural network approaches.

8.1 Final Recommendations

- **Daily forecasting:** Deploy Random Forest with the full feature set; retrain periodically as new data becomes available.
- **15-minute forecasting:** Deploy XGBoost with lag and cyclical time features; consider online learning for real-time adaptation.

9 Future Work

The following directions are proposed to further enhance the performance, robustness, and applicability of the forecasting framework:

1. **Incorporation of exogenous variables:** Electricity demand is strongly influenced by external factors such as temperature, humidity, and economic activity. Integrating weather data and other external indicators as exogenous features is expected to significantly improve model performance.
2. **Advanced feature engineering:** Additional temporal features such as longer lag windows, multi-scale rolling statistics, and interaction terms can be explored to better capture complex demand dynamics across both datasets.
3. **Hyperparameter optimisation:** The current models use manually selected parameters. Systematic optimisation techniques such as grid search, random search, or Bayesian optimisation can be employed to identify optimal configurations and further improve predictive accuracy.
4. **Enhanced deep learning models:** While linear deep learning models (DLinear, NLinear) were explored, more advanced architectures such as LSTM, GRU, Temporal Fusion Transformer (TFT), and PatchTST can be implemented with proper tuning and longer training to capture long-range dependencies and complex temporal patterns.
5. **Hybrid modelling approaches:** Combining classical time series models with machine learning methods (e.g., modelling residuals from SARIMA using XGBoost) may help leverage both statistical interpretability and non-linear modelling capabilities.
6. **Model deployment and automation:** Developing a real-time forecasting pipeline with automated data ingestion, model retraining, and performance monitoring would enable practical deployment of the proposed system.
7. **Scalability and regional analysis:** Extending the framework to region-wise or state-level demand forecasting can provide more granular insights and support localized energy planning and management.

A Model Hyperparameter Summary

Table 9: Hyperparameters Used for ML and DL Models

Model	**Hyperparameters**
XGBoost (Daily)	n_estimators=300, learning_rate=0.05, max_depth=5, subsample=0.8, colsample_bytree=0.8, random_state=42
XGBoost (Final, Daily)	n_estimators=200, learning_rate=0.05, max_depth=6, subsample=0.8, colsample_bytree=0.8, random_state=42
XGBoost (15-min)	n_estimators=150, learning_rate=0.05, max_depth=5, subsample=0.8, colsample_bytree=0.8, random_state=42
XGBoost (Final, 15-min)	n_estimators=200, learning_rate=0.05, max_depth=6, random_state=42
Random Forest (Daily)	n_estimators=200, max_depth=10, n_jobs=-1, random_state=42
Random Forest (15-min)	n_estimators=150, max_depth=10, n_jobs=-1, random_state=42
Random Forest (Final, 15-min)	n_estimators=100, n_jobs=-1, random_state=42
DLinear / NLinear	window=14 (daily) or 96 (15-min), epochs=20, lr=0.01, Adam optimiser, MSE loss
SARIMA / SARIMAX	order=(2,0,2), seasonal_order=(1,1,1,7), enforce_stationarity=False